

Supervisory Learning Bot — Product Specification

1. Product Background

An AI focus learning robot built around the Pomodoro Technique, guided by child attention development theory. It uses visual perception, multimodal interaction, and intelligent guidance to provide real-time supervision of the learning process, dynamic feedback, and long-term habit formation. The core function is supervising children while they do homework and study.

Problems it solves:

1. Parents can't stay with children continuously to monitor them, making it hard to catch distraction in real time.
2. Children have weak self-management skills and easily space out, get distracted, play with their phone, or leave their seat frequently.
3. Traditional Pomodoro tools only offer time management — they lack real-time awareness of learning state and dynamic guidance.
4. Parents can only see learning outcomes, not how focus levels and behavior changed throughout the session.

The final product aims to be an AI learning companion system with a child-side and parent-side component used together.

The child-side comes in two forms:

Hardware learning bot placed on the desk — uses a camera, microphone, speaker, and display screen to accompany the child while studying.

Lightweight mini-program or App that uses the phone or tablet camera to run Pomodoro sessions, detect learning state, and deliver AI reminders.

While the child studies, the system manages learning rhythm via the Pomodoro flow, detecting whether the child is seated, doing homework, spacing out, using a phone, or slumping with fatigue — and responds with voice prompts, expressions, or an animated avatar.

The parent-side is a mini-program or App showing a study dashboard: whether the child is currently focused, how long they've been studying, how many Pomodoro rounds they've completed today, how many times they got distracted, and how focus has trended over time.

2. Software Modules

Feature	Definition
Real-time state recognition	Continuously detects: in-seat/out-of-seat, looking at desk/distracted, writing activity, posture, phone use, fatigue risk
Pomodoro flow	Prepare, Focus, Break, Resume, Complete
Rules engine	Events and thresholds trigger corresponding actions
Script generation	Rules provide intent; the agent generates the final spoken script
Voice output (TTS)	Converts scripts to audio

Facial expression state machine	Generates expressions based on current action and state machine state
Parent-side display	View focus duration, interruption events, posture trends, session summaries
Data storage & privacy	Local, minimal storage; parent authorization required before use; history can be deleted

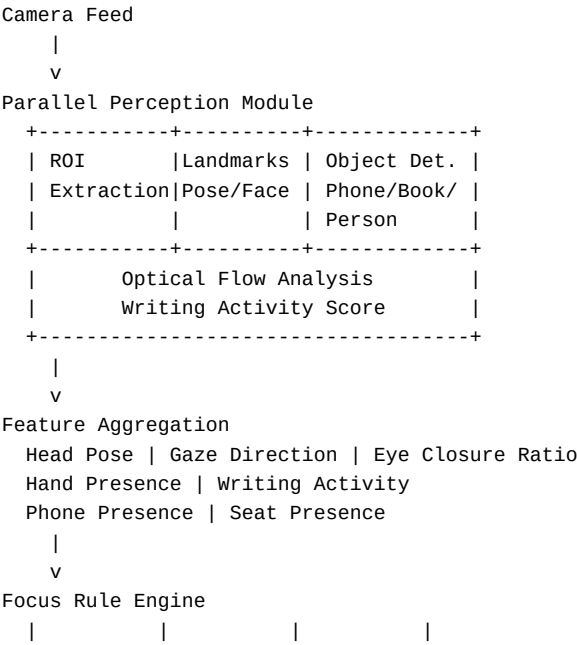
3. Flow

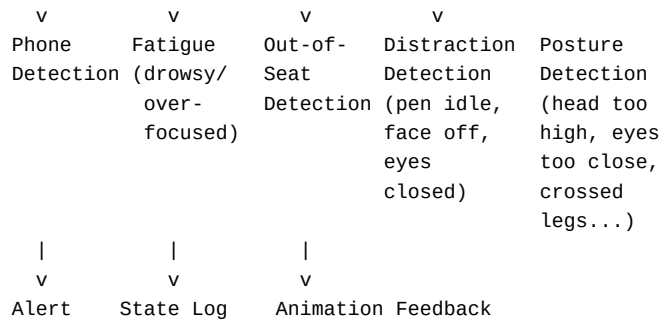
Real-time recognition → Pomodoro state machine → Rules engine → Script generation → TTS → TFT expression feedback → Parent-side display

Pomodoro State Machine

State	Entry Condition	Actions	Exit Condition
IDLE	User hasn't started	Show standby expression, wait for start	Tap start or voice wake
PREPARE	1–2 min before study	Guide child to sit, prep stationery, confirm desk ROI	Child ready or prep time expires
FOCUS	Entering focus phase	Timer running, recognition active, light feedback	Focus duration reached or serious interruption
BREAK	Normal rest	Relaxation script, rest expression, disable intervention	Break duration expires
RESUME_CHECK	Post-break wrap-up	Check if child returned to desk	Back at desk → re-enter FOCUS
COMPLETE	Round/session ends	Summary script, parent-side digest	Session ends

Focus Phase Flow





Break Phase Flow

